

STORM: Simplicial Topological Recurrent Model for Dynamics on Higher-Order Domains

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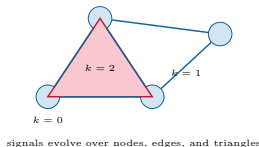
Talk Overview

- ▶ Motivation and Gap
- ▶ STORM Architecture
- ▶ Theoretical Properties
- ▶ Numerical Results
- ▶ Conclusion

Multivariate Time Series on Higher-Order Topologies

- ▶ We consider **multivariate time series** that live on **higher-order topologies**.
- ▶ In this work, the topology is represented by a simplicial complex:

$k = 0$: nodes, $k = 1$: edges, $k = 2$: filled triangles.



- ▶ Examples:
 - ⇒ traffic: sensors, road interactions, local traffic regions
 - ⇒ water networks: pressures on nodes, flows on pipes
 - ⇒ neural activity: neurons, co-activity pairs, higher-order co-activity motifs
- ▶ **Goal:** model time series whose dependencies are both temporal and higher-order topological.

Topological Time Series

- ▶ We consider a **fixed simplicial complex** $\mathcal{K}$.
- ▶ Input: a collection of time series over simplex orders

$$\left\{ \mathbf{X}^{(k)} = [\mathbf{x}_1^{(k)}, \dots, \mathbf{x}_T^{(k)}] \in \mathbb{R}^{N_k \times T} \right\}_{k=0}^K.$$

- ▶ **Goal:** learn a topology-aware temporal representation

$$\left\{ \mathbf{x}_1^{(k)}, \dots, \mathbf{x}_t^{(k)} \right\}_{k=0}^K \longrightarrow \left\{ \mathbf{h}_t^{(k)} \right\}_{k=0}^K.$$

- ▶ This representation can be used for downstream tasks such as forecasting, anomaly detection, and missing-data imputation.

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- ▶ **Sequential models** process time, but ignore the topology:

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- ▶ **Disjoint topological-temporal models** first encode topology, then apply a temporal model:

topological encoder $\longrightarrow$ RNN.

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topological encoder $\longrightarrow$ RNN.

- ▶ **Gap:** topology should structure how the temporal state evolves.

STORM: Simplicial Topological Recurrent Model

- ▶ STORM combines two ideas:

- ⇒ **gated memory** from LSTM/xLSTM-type recurrent cells

- ⇒ **topological filters** inside the recurrent gates

topological time series $\longrightarrow$ **topology-aware gates** $\longrightarrow$ **topological memory** $\longrightarrow$ **representation**

- ▶ In STORM, the topology controls

what is written, what is forgotten, what is read out.

- ▶ **Main idea:** STORM puts simplicial filters inside the gates of a recurrent memory model.

STORM States and Recurrent Update

- ▶ STORM maintains hidden and cell states on each simplex order, of a fixed simplicial complex $\mathcal{K}$.

$$\mathbf{X}_t^{(k)} \in \mathbb{R}^{N_k \times d_k}, \quad \mathbf{h}_t^{(k)}, \mathbf{c}_t^{(k)} \in \mathbb{R}^{N_k \times H}, \quad k = 0, \dots, K.$$

- ▶ For each gate $g \in \{i, z, o, \tau\}$, compute a topological pre-activation $\mathbf{G}_g^{(k)}$.

$$\mathbf{i}_t^{(k)} = \sigma(\mathbf{G}_i^{(k)}), \quad \tilde{\mathbf{c}}_t^{(k)} = \tanh(\mathbf{G}_z^{(k)}), \quad \mathbf{o}_t^{(k)} = \sigma(\mathbf{G}_o^{(k)}).$$

- ▶ STORM uses a positive timescale and exponential forget gate:

$$\boldsymbol{\tau}_t^{(k)} = \text{softplus}(\mathbf{G}_\tau^{(k)}), \quad \mathbf{f}_t^{(k)} = \exp(-\Delta t \oslash \boldsymbol{\tau}_t^{(k)}).$$

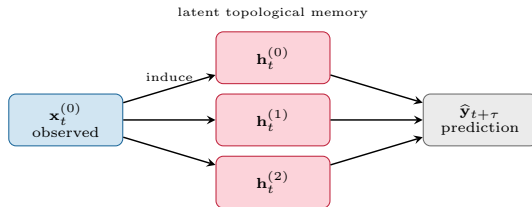
- ▶ The memory update is LSTM-like:

$$\mathbf{c}_t^{(k)} = \mathbf{f}_t^{(k)} \odot \mathbf{c}_{t-1}^{(k)} + \mathbf{i}_t^{(k)} \odot \tilde{\mathbf{c}}_t^{(k)}, \quad \mathbf{h}_t^{(k)} = \mathbf{o}_t^{(k)} \odot \tanh(\mathbf{c}_t^{(k)}).$$

- ▶ **Main idea:** STORM stores and updates memory on every simplex order.

Higher-Order Structure May Be Latent

- ▶ In many applications, the observed signal is only on nodes $\mathbf{y}_t = \mathbf{x}_t^{(0)}$.
- ▶ However, the hidden memory can still live on all simplex orders $\mathbf{h}_t^{(0)}, \mathbf{h}_t^{(1)}, \dots, \mathbf{h}_t^{(K)}$.



- ▶ The higher-order latent space can be **induced by the observed signals** and then processed with higher-order topological filters.

STORM Gates Are Topological

- The pre-activation of each gate $g \in \{i, z, o, \tau\}$ is computed from two branches:

$$\mathbf{G}_g^{(k)} = \mathbf{G}_{X,g}^{(k)} + \mathbf{G}_{h,g}^{(k)} + \mathbf{1}_{N_k} (\mathbf{b}_g^{(k)})^\top.$$

- **Input branch:**

$$\mathbf{G}_{X,g}^{(k)} = \Phi_{X,g}^{(k)} \left(\{\mathbf{X}_t^{(j)}\}_{j=0}^K; \mathcal{K} \right).$$

- **Hidden branch:**

$$\mathbf{G}_{h,g}^{(k)} = \Phi_{h,g}^{(k)} \left(\{\mathbf{h}_{t-1}^{(j)}\}_{j=0}^K; \mathcal{K} \right).$$

- The functions $\Phi_{X,g}^{(k)}$ and $\Phi_{h,g}^{(k)}$ are simplicial convolutional filters with cross-order injections.

Same- and Cross-Order Topology

- For each gate and order, STORM combines two kinds of topological information.

$$\mathbf{G}_{X,g}^{(k)} = \underbrace{\text{intra-order filtering of } \mathbf{X}_t^{(k)}}_{\mathbf{L}_k^\downarrow, \mathbf{L}_k^\uparrow} + \underbrace{\text{cross-order injection from } k-1, k+1}_{\mathbf{B}_k, \mathbf{B}_{k+1}}.$$

- In expanded form:

$$\begin{aligned} \mathbf{G}_{X,g}^{(k)} = & \mathbf{X}_t^{(k)} \boldsymbol{\Theta}_{g,\text{id}}^{(k)} + \sum_{p=1}^P (\mathbf{L}_k^\downarrow)^p \mathbf{X}_t^{(k)} \boldsymbol{\Theta}_{g,\downarrow,p}^{(k)} + \sum_{p=1}^P (\mathbf{L}_k^\uparrow)^p \mathbf{X}_t^{(k)} \boldsymbol{\Theta}_{g,\uparrow,p}^{(k)} \\ & + \sum_{p=0}^P (\mathbf{L}_k^\downarrow)^p \tilde{\mathbf{X}}_{t,\downarrow}^{(k)} \boldsymbol{\Theta}_{g,\text{inj},\downarrow,p}^{(k)} + \sum_{p=0}^P (\mathbf{L}_k^\uparrow)^p \tilde{\mathbf{X}}_{t,\uparrow}^{(k)} \boldsymbol{\Theta}_{g,\text{inj},\uparrow,p}^{(k)} \end{aligned}$$

- Cross-order injected signals are obtained through the boundary maps:

$$\tilde{\mathbf{X}}_{t,\downarrow}^{(k)} = \mathbf{B}_k^\top \mathbf{X}_t^{(k-1)}, \quad \tilde{\mathbf{X}}_{t,\uparrow}^{(k)} = \mathbf{B}_{k+1} \mathbf{X}_t^{(k+1)}.$$

- The hidden branch is analogous, replacing $\mathbf{X}_t$ by $\mathbf{h}_{t-1}$.

STORM Injection Modes

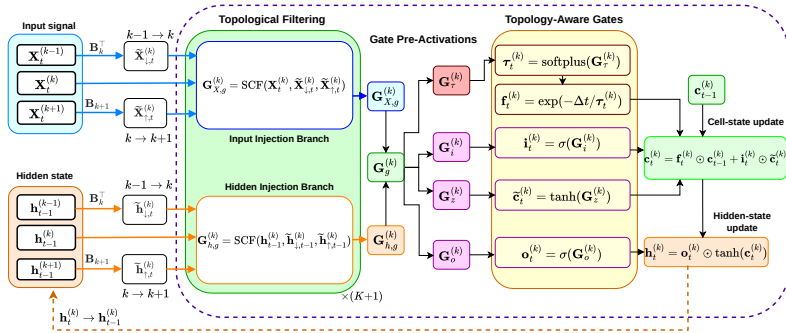
- ▶ STORM injection modes control **where cross-order interactions occur**, not whether a branch is used.
- ▶ In all modes, both branches contribute to each gate pre-activation:

$$\mathbf{G}_g^{(k)} = \mathbf{G}_{X,g}^{(k)} + \mathbf{G}_{h,g}^{(k)} + \mathbf{1}_{N_k} (\mathbf{b}_g^{(k)})^\top.$$

Variant	Input branch $\mathbf{X}_t$	Hidden branch $\mathbf{h}_{t-1}$
STORM-input	intra-order + cross-order	intra-order only
STORM-hidden	intra-order only	intra-order + cross-order
STORM-both	intra-order + cross-order	intra-order + cross-order

- ▶ **Cross-order injection** is switched on in the input branch, hidden branch, or both.

STORM Architecture



- Two parallel SCF branches process the current input $\mathbf{X}_t^{(k)}$ and previous hidden state $\mathbf{h}_{t-1}^{(k)}$.
- Cross-order signals from $k-1$ and $k+1$ are injected through boundary operators.
- The resulting gate pre-activations drive the recurrent update for $g \in \{\tau, i, z, o\}$.

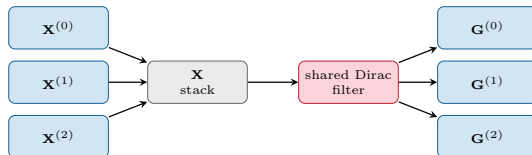
LightSTORM: Shared Dirac Filters

- ▶ STORM uses order-specific simplicial filters $\left\{ \mathbf{H}^{(k)} \left(\mathbf{L}_k^\downarrow, \mathbf{L}_k^\uparrow \right) \right\}_{k=0}^K$.
- ▶ LightSTORM stacks all orders and processes them with shared Dirac-polynomial filters.

$$\mathbf{X} = \text{stack} \left(\mathbf{X}^{(0)}, \dots, \mathbf{X}^{(K)} \right) \in \mathbb{R}^{N \times H}, \quad N = \sum_{k=0}^K N_k.$$

$$\mathbf{G}_{X,g} = \mathbf{X} \boldsymbol{\Theta}_{g,0} + \sum_{p=1}^J \left(\mathbf{D}^{(d)} \right)^p \mathbf{X} \boldsymbol{\Theta}_{g,p}^{(d)} + \sum_{p=1}^J \left(\mathbf{D}^{(u)} \right)^p \mathbf{X} \boldsymbol{\Theta}_{g,p}^{(u)}.$$

- ▶ The hidden branch is analogous, replacing $\mathbf{X}$ with $\mathbf{h}$.



coefficients shared across orders

Permutation Equivariance

- ▶ Let $\mathbf{P}_k \in \{0, 1\}^{N_k \times N_k}$ be a permutation matrix relabeling the k -simplices.
- ▶ Relabel the complex and the input signals consistently:

$$\mathbf{B}'_k = \mathbf{P}_{k-1} \mathbf{B}_k \mathbf{P}_k^\top, \quad \mathbf{X}_t^{(k)'} = \mathbf{P}_k \mathbf{X}_t^{(k)}.$$

Proposition: If STORM is runs on the relabeled complex with relabeled initial states, then for all $t \geq 0$ and $k = 0, \dots, K$,

$$\mathbf{h}_t^{(k)'} = \mathbf{P}_k \mathbf{h}_t^{(k)}, \quad \mathbf{c}_t^{(k)'} = \mathbf{P}_k \mathbf{c}_t^{(k)}.$$

- ▶ $\mathbf{h}_t^{(k)}$ and $\mathbf{c}_t^{(k)}$ are the hidden and cell states on k -simplices.
- ▶ **Meaning:** STORM depends on the topology, not on the arbitrary ordering of nodes, edges, or triangles.

Stability to Topology Perturbations

- ▶ $\mathcal{K}$ be the original complex and $\widehat{\mathcal{K}}$ a perturbed version
- ▶ The boundary operators are perturbed as

$$\widehat{\mathbf{B}}_k = \mathbf{B}_k + \Delta\mathbf{B}_k, \quad \|\Delta\mathbf{B}_k\|_2 \leq \eta r_k.$$

- ▶ η is the perturbation size and r_k is the scale of the k -th boundary operator

Theorem: Under **bounded-filter** and **contractive-memory** conditions, running STORM on $\mathcal{K}$ and $\widehat{\mathcal{K}}$ yields constants $C_{\text{BIBO}}, C'_{\text{BIBO}} < \infty$ such that

$$\|\mathbf{Y}_t - \widehat{\mathbf{Y}}_t\| \leq C_{\text{BIBO}}\eta + C'_{\text{BIBO}}\eta^2, \quad t \geq 1.$$

- ▶ **Meaning:** small topology perturbations induce controlled output changes, uniformly over time.

Generalization Bound

- ▶ Let $f_\theta \in \mathcal{F}_T$ be a STORM predictor trained on M trajectories $\{(\mathbf{X}_{1:T}^{(m)}, \mathbf{Y}^{*(m)})\}_{m=1}^M$, and let $R(f_\theta)$ and $\hat{R}_M(f_\theta)$ denote the true and empirical risks.

Theorem: If the recurrent state class **contracts in Rademacher complexity** with factor $q_{\text{gen}} < 1$, then with probability at least $1 - \delta$, uniformly over $\theta \in \Theta$,

$$R(f_\theta) \leq \hat{R}_M(f_\theta) + \frac{2L_\ell c_{\text{out}} \mathfrak{C}_{\text{full}}}{(1 - q_{\text{gen}})\sqrt{M}} + B_\ell \sqrt{\frac{\log(2/\delta)}{2M}}.$$

$$\mathfrak{C}_{\text{full}} := C_{\text{gate}}(K + 1)(4P + 3)[W_{\text{max}}\bar{r}(B_X + B_S) + b_{\text{max}}],$$

$$\bar{r} := \max\left(1, \max_k r_{k,\downarrow}, \max_k r_{k,\uparrow}\right).$$

- ▶ L_ℓ, B_ℓ : loss Lipschitz constant and bound; c_{out} : readout Lipschitz constant; P : filter order; $K + 1$: number of simplex orders.
- ▶ $W_{\text{max}}, b_{\text{max}}$: bounds on filter matrices and biases; B_X, B_S : bounds on inputs and recurrent states; $\bar{r}$: bound on cross-order operator norms.
- ▶ C_{gate} : universal constant collecting fixed Lipschitz constants of gate nonlinearities and Hadamard products.

Numerical Setup

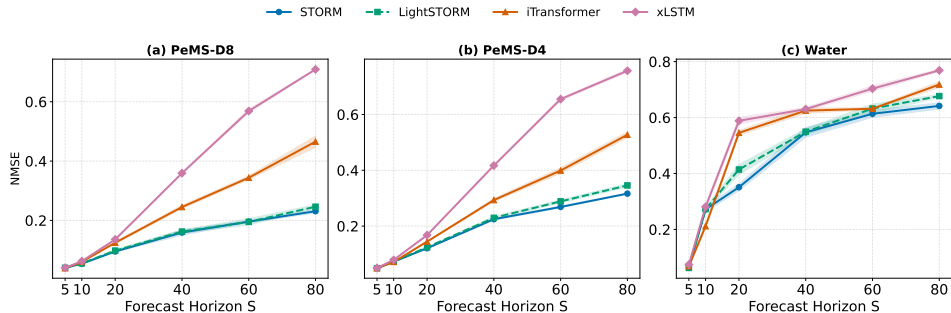
- We evaluate multi-order forecasting over fixed simplicial complexes:

$$\{\mathbf{X}_1^{(k)}, \dots, \mathbf{X}_t^{(k)}\}_{k=0}^K \longrightarrow \{\widehat{\mathbf{X}}_{t+1:t+S}^{(k)}\}_{k=0}^K, \quad S \in \{5, 10, 20, 40, 60, 80\}.$$

Dataset	T	N_0	N_1	N_2	Signals
PeMS-D4	16,992	307	340	29	node traffic; lifted edge/triangle signals
PeMS-D8	17,856	170	274	80	node traffic; lifted edge/triangle signals
Water	4,480	36	40	0	node pressure; edge flow

- **Metric:** final-step pooled NMSE at horizon S , lower is better.
- **Baselines:**
 - ⇒ topology-unaware: LSTM, xLSTM, Transformer, iTransformer
 - ⇒ graph-based: GCN, GCN-xLSTM
 - ⇒ simplicial: SC-VAR, SCRNN

Main Forecasting Results



NMSE vs. forecasting step S (mean ± 1 std over 5 seeds; lower is better). For STORM and LightSTORM, each point reports the injection variant with the lowest mean NMSE at the corresponding dataset and horizon.

- **Take-home:** gains are largest at medium and long horizons on the traffic networks and become consistent from medium horizons onward on Water.

Role of Topology

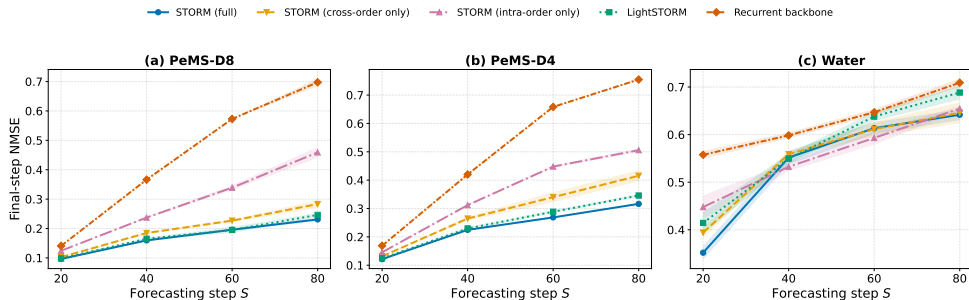


Figure: Topology ablation. Final-step NMSE at forecasting steps $S \in \{20, 40, 60, 80\}$. Lower is better. Full STORM combines intra-order Hodge filtering with cross-order boundary coupling.

- **Take-home:** topology-aware variants separate more clearly from the topology-free recurrent backbone at longer horizons.

Injection Modes: Results

Table: Injection-mode ablation on PeMS-D8. Lower is better. Horizontal separators distinguish input, hidden, and both injection modes.

Model	Mode	H	$S=10$	$S=40$	$S=80$
STORM	input	16	0.0616 ± 0.0006	0.2956 ± 0.0047	0.5655 ± 0.0088
	input	32	0.0587 ± 0.0007	0.2233 ± 0.0083	0.3913 ± 0.0053
	hidden	16	0.0578 ± 0.0014	0.1908 ± 0.0018	0.3256 ± 0.0200
	hidden	32	0.0560 ± 0.0004	0.1592 ± 0.0093	0.2370 ± 0.0125
	both	16	0.0573 ± 0.0001	0.1880 ± 0.0110	0.3376 ± 0.0098
	both	32	0.0541 ± 0.0007	0.1596 ± 0.0041	0.2310 ± 0.0059
LightSTORM	input	16	0.0636 ± 0.0004	0.3570 ± 0.0057	0.6919 ± 0.0100
	input	32	0.0626 ± 0.0008	0.3105 ± 0.0066	0.5867 ± 0.0046
	hidden	16	0.0588 ± 0.0001	0.2123 ± 0.0043	0.4065 ± 0.0064
	hidden	32	0.0552 ± 0.0008	0.1624 ± 0.0055	0.2511 ± 0.0177
	both	16	0.0600 ± 0.0006	0.2155 ± 0.0106	0.3736 ± 0.0104
	both	32	0.0558 ± 0.0006	0.1652 ± 0.0037	0.2462 ± 0.0067

Per-Order Results: PeMS-D8

Model	$S=40$			$S=80$		
	Nodes	Edges	Triangles	Nodes	Edges	Triangles
STORM (both)	<u>0.1600</u>	<u>0.1622</u>	0.1496	0.2272	0.2379	0.2153
STORM (hidden)	0.1598	0.1612	<u>0.1510</u>	<u>0.2316</u>	<u>0.2439</u>	<u>0.2244</u>
LightSTORM (both)	0.1630	0.1686	0.1584	0.2352	0.2569	<u>0.2244</u>
LightSTORM (hidden)	0.1609	0.1659	0.1532	0.2438	0.2627	0.2265
iTransformer	0.2440	0.2460	0.2443	0.4488	0.4742	0.4718
xLSTM	0.3466	0.3646	0.3681	0.6621	0.7303	0.7374
GCN-xLSTM	0.2371	0.3104	0.3391	0.4453	0.6063	0.6627
SC-VAR	0.5220	0.5840	0.6080	0.7946	0.9392	0.9483

- **Take-home:** at $S = 80$, STORM gives the strongest per-order results across nodes, edges, and triangles.

Conclusion

- ▶ We proposed STORM and LightSTORM
 - ⇒ STORM places simplicial filters inside recurrent gates
 - ⇒ LightSTORM uses shared Dirac filters for a lighter topological recurrent model
- ▶ Theoretical properties
 - ⇒ permutation equivariance to simplex relabeling
 - ⇒ stability to small topology perturbations
 - ⇒ generalization control under temporal complexity contraction
- ▶ Numerical take-home
 - ⇒ topology-aware memory improves medium- and long-horizon forecasting
 - ⇒ gains are strongest on richer complexes with nodes, edges, and triangles
 - ⇒ topology and injection-mode ablations support the role of higher-order memory

Thank you for the attention.

Questions?

Contact

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LinkedIn QR code

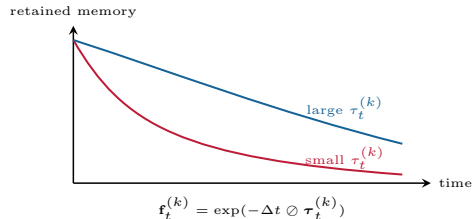
Backup Slides

STORM: Exponential Forget Gate

- STORM uses a positive, topology-dependent timescale for the forget gate.

$$\tau_t^{(k)} = \text{softplus} \left(\mathbf{G}_\tau^{(k)} \right), \quad \mathbf{f}_t^{(k)} = \exp \left(-\Delta t \oslash \tau_t^{(k)} \right).$$

- The cell update remains LSTM-like: $\mathbf{c}_t^{(k)} = \mathbf{f}_t^{(k)} \odot \mathbf{c}_{t-1}^{(k)} + \mathbf{i}_t^{(k)} \odot \tilde{\mathbf{c}}_t^{(k)}$.



- A small timescale forgets quickly; a large timescale preserves memory longer.
- Main idea:** STORM learns how fast each topological memory component should decay.

Permutation Equivariance

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- ▶ Relabel the complex and the input signals consistently:

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- ▶ $\mathbf{h}_t^{(k)}$ and $\mathbf{c}_t^{(k)}$ are the hidden and cell states on k -simplices.
- ▶ **Meaning:** STORM depends on the topology, not on the arbitrary ordering of nodes, edges, or triangles.

- ▶ **Orientation:** reversing orientations introduces sign flips in oriented signals and boundary matrices
- ▶ Hodge and boundary filters are orientation equivariant

Remark: STORM is **not** orientation equivariant, **because** the sigmoid, softplus, clamp, and exponential gate **nonlinearities** do **not commute** with sign flips.

- ▶ **Odd nonlinearities are insufficient:** input, forget, and output gates multiply sign-changing states, and the gate values would need to remain orientation invariant

Stability to Topology Perturbations

- ▶ $\mathcal{K}$ be the original complex and $\hat{\mathcal{K}}$ a perturbed version
- ▶ The boundary operators are perturbed as

$$\hat{\mathbf{B}}_k = \mathbf{B}_k + \Delta\mathbf{B}_k, \quad \|\Delta\mathbf{B}_k\|_2 \leq \eta r_k.$$

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- ▶ Let $f_\theta \in \mathcal{F}_T$ be a STORM predictor trained on M trajectories $(\mathbf{X}_{1:T}^{(m)}, \mathbf{Y}^{*(m)})$.
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- ▶ C_{gate} : universal constant collecting the fixed Lipschitz constants of gate nonlinearities and Hadamard products.

Numerical Setup: Forecasting Task

- ▶ We evaluate multivariate time-series forecasting over fixed simplicial complexes.
- ▶ At time t , the model observes signals on multiple simplex orders and predicts

$$\widehat{\mathbf{X}}_{t+1:t+S}^{(k)} = \mathbf{h}_t^{(k)} \mathbf{W}_{\text{out}}^{(k)} + \mathbf{b}_{\text{out}}^{(k)}, \quad k = 0, \dots, K.$$

- ▶ The decoder outputs all S forecast steps in one forward pass, with

$$S \in \{5, 10, 20, 40, 60, 80\}.$$

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Water	4,480	36	40	0	node pressure; edge flow

- ▶ **Main idea:** forecast node, edge, and triangle signals jointly whenever available.

Numerical Setup: Loss, Metrics, Baselines

- ▶ Training uses a balanced forecasting loss, averaging MSE across simplex orders:

$$\mathcal{L}_{\text{balanced}} = \frac{1}{K+1} \sum_{k=0}^K \mathcal{L}^{(k)}.$$

- ▶ Evaluation uses final-step pooled NMSE at horizon S :

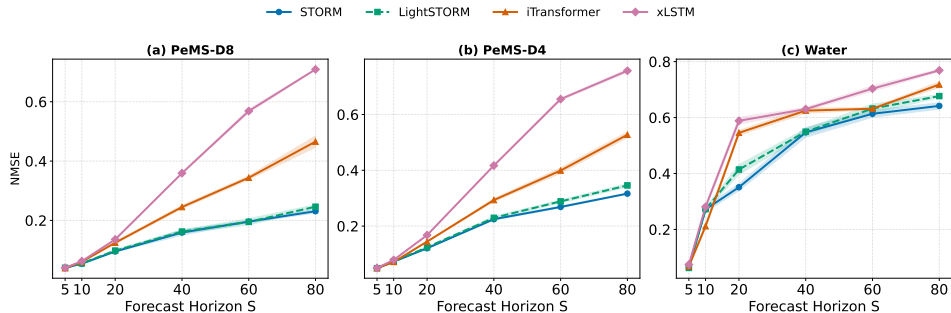
$$\text{NMSE}_S = \frac{\sum_{k,t,i,j} \left(\hat{Y}_{t,i,S,j}^{(k)} - Y_{t,i,S,j}^{(k)} \right)^2}{\sum_{k,t,i,j} \left(Y_{t,i,S,j}^{(k)} - \bar{Y}_S^{(k)} \right)^2}.$$

Lower is better.

- ▶ **Baselines**

- ⇒ Topology-unaware: LSTM, xLSTM, Transformer, iTransformer
- ⇒ Graph-based: GCN, GCN-xLSTM
- ⇒ Simplicial: SC-VAR, SCRNN
- ⇒ Ours: STORM, LightSTORM

Main Forecasting Results



NMSE vs. forecasting step S (mean ± 1 std over 5 seeds; lower is better). For STORM and LightSTORM, each point reports the injection variant with the lowest mean NMSE at the corresponding dataset and horizon.

- **Take-home:** gains are largest at medium and long horizons on the traffic networks and become consistent from medium horizons onward on Water.

Main Results: PeMS-D8

Table: Final-step pooled NMSE on PeMS-D8. Lower is better. Bold indicates best result and underline second best for each horizon.

Model	$S=5$	$S=10$	$S=20$	$S=40$	$S=60$	$S=80$
STORM (both)	0.0404 ± 0.0005	0.0541 ± 0.0007	<u>0.0963 ± 0.0004</u>	<u>0.1596 ± 0.0041</u>	<u>0.1959 ± 0.0033</u>	0.2310 ± 0.0059
STORM (hidden)	0.0402 ± 0.0005	0.0560 ± 0.0004	0.0953 ± 0.0019	0.1592 ± 0.0093	0.2008 ± 0.0065	<u>0.2370 ± 0.0125</u>
LightSTORM (both)	0.0403 ± 0.0003	0.0558 ± 0.0006	0.0982 ± 0.0037	0.1652 ± 0.0037	0.1951 ± 0.0105	0.2462 ± 0.0067
LightSTORM (hidden)	0.0403 ± 0.0004	<u>0.0552 ± 0.0008</u>	0.0983 ± 0.0029	0.1624 ± 0.0055	0.2063 ± 0.0090	0.2511 ± 0.0177
iTransformer	0.0388 ± 0.0001	0.0604 ± 0.0005	0.1248 ± 0.0038	0.2451 ± 0.0055	0.3439 ± 0.0062	0.4656 ± 0.0188
xLSTM	0.0403 ± 0.0003	0.0625 ± 0.0006	0.1359 ± 0.0022	0.3593 ± 0.0016	0.5687 ± 0.0065	0.7092 ± 0.0009
Transformer	0.0426 ± 0.0007	0.0653 ± 0.0012	0.1427 ± 0.0033	0.3726 ± 0.0087	0.5720 ± 0.0033	0.6879 ± 0.0061
LSTM	<u>0.0392 ± 0.0004</u>	0.0622 ± 0.0009	0.1373 ± 0.0007	0.3616 ± 0.0085	0.5640 ± 0.0064	0.6958 ± 0.0066
GCN-xLSTM	0.0404 ± 0.0002	0.0622 ± 0.0006	0.1333 ± 0.0029	0.2909 ± 0.0064	0.4427 ± 0.0053	0.5623 ± 0.0092
SC-VAR	0.0508 ± 0.0002	0.0966 ± 0.0007	0.2407 ± 0.0003	0.5675 ± 0.0004	0.7904 ± 0.0001	0.8934 ± 0.0007
SCRNN	0.8706 ± 0.0004	0.8726 ± 0.0003	0.8753 ± 0.0006	0.8817 ± 0.0060	0.8820 ± 0.0033	0.8856 ± 0.0020
<i>Gain vs. best baseline</i>	-3.6%	+10.4%	+23.6%	+35.0%	+43.3%	+50.4%

- **Take-home:** STORM gives the strongest long-horizon results, with the largest gain at $S = 80$.

Main Results: PeMS-D4

Table: Final-step pooled NMSE on PeMS-D4. Lower is better. Bold indicates best result and underline second best for each horizon.

Model	$S=5$	$S=10$	$S=20$	$S=40$	$S=60$	$S=80$
STORM (both)	0.0495 ± 0.0002	<u>0.0725 ± 0.0004</u>	<u>0.1214 ± 0.0013</u>	0.2247 ± 0.0033	0.2686 ± 0.0031	0.3164 ± 0.0017
STORM (hidden)	<u>0.0494 ± 0.0003</u>	0.0728 ± 0.0008	0.1206 ± 0.0020	<u>0.2259 ± 0.0006</u>	<u>0.2759 ± 0.0077</u>	<u>0.3287 ± 0.0179</u>
LightSTORM (both)	0.0494 ± 0.0004	0.0735 ± 0.0005	0.1244 ± 0.0008	0.2297 ± 0.0051	0.2886 ± 0.0051	0.3458 ± 0.0061
LightSTORM (hidden)	0.0491 ± 0.0002	0.0729 ± 0.0009	0.1253 ± 0.0023	0.2449 ± 0.0069	0.3093 ± 0.0133	0.3626 ± 0.0074
iTransformer	<u>0.0494 ± 0.0001</u>	0.0715 ± 0.0005	0.1444 ± 0.0012	0.2933 ± 0.0054	0.3991 ± 0.0097	0.5274 ± 0.0100
xLSTM	0.0498 ± 0.0002	0.0788 ± 0.0005	0.1676 ± 0.0014	0.4170 ± 0.0015	0.6550 ± 0.0042	0.7567 ± 0.0060
Transformer	0.0516 ± 0.0002	0.0829 ± 0.0003	0.1740 ± 0.0029	0.4298 ± 0.0039	0.6640 ± 0.0065	0.7605 ± 0.0018
LSTM	0.0499 ± 0.0002	0.0789 ± 0.0004	0.1690 ± 0.0005	0.4202 ± 0.0036	0.6500 ± 0.0052	0.7589 ± 0.0044
GCN-xLSTM	0.0495 ± 0.0004	0.0779 ± 0.0004	0.1573 ± 0.0012	0.3641 ± 0.0022	0.5420 ± 0.0043	0.6453 ± 0.0077
SC-VAR	0.0623 ± 0.0004	0.1173 ± 0.0009	0.2717 ± 0.0005	0.6046 ± 0.0004	0.8303 ± 0.0002	0.9339 ± 0.0004
SCRNN	0.8036 ± 0.0002	0.8057 ± 0.0012	0.8137 ± 0.0010	0.8187 ± 0.0023	0.8151 ± 0.0013	0.8257 ± 0.0011
<i>Gain vs. best baseline</i>	<i>+0.6%</i>	<i>−1.4%</i>	<i>+16.5%</i>	<i>+23.4%</i>	<i>+32.7%</i>	<i>+40.0%</i>

- **Take-home:** STORM is competitive at short horizons and improves over baselines at longer horizons.

Main Results: Water

Table: Final-step pooled NMSE on Water. Lower is better. Bold indicates best result and underline second best for each horizon.

Model	$S=5$	$S=10$	$S=20$	$S=40$	$S=60$	$S=80$
STORM (both)	0.0679 ± 0.0010	0.2772 ± 0.0035	<u>0.3519 ± 0.0144</u>	0.5517 ± 0.0143	0.6135 ± 0.0125	0.6417 ± 0.0113
STORM (hidden)	0.0655 ± 0.0027	0.2720 ± 0.0098	0.3509 ± 0.0122	0.5471 ± 0.0176	<u>0.6222 ± 0.0123</u>	<u>0.6548 ± 0.0105</u>
LightSTORM (both)	0.0697 ± 0.0010	0.3016 ± 0.0061	0.4143 ± 0.0175	<u>0.5495 ± 0.0103</u>	0.6379 ± 0.0097	0.6884 ± 0.0144
LightSTORM (hidden)	0.0696 ± 0.0020	0.3054 ± 0.0053	0.4144 ± 0.0227	0.5604 ± 0.0077	0.6321 ± 0.0163	0.6768 ± 0.0055
iTransformer	0.0683 ± 0.0021	0.2118 ± 0.0075	0.5455 ± 0.0085	0.6249 ± 0.0095	0.6315 ± 0.0033	0.7180 ± 0.0104
xLSTM	0.0745 ± 0.0018	0.2809 ± 0.0088	0.5884 ± 0.0123	0.6303 ± 0.0060	0.7035 ± 0.0098	0.7687 ± 0.0046
Transformer	0.1012 ± 0.0010	0.3400 ± 0.0230	0.5490 ± 0.0029	0.6091 ± 0.0054	0.6654 ± 0.0052	0.7073 ± 0.0062
LSTM	<u>0.0658 ± 0.0039</u>	<u>0.2380 ± 0.0083</u>	0.5576 ± 0.0081	0.5984 ± 0.0061	0.6469 ± 0.0065	0.7090 ± 0.0088
GCN-xLSTM	0.0739 ± 0.0041	0.3252 ± 0.0207	0.5305 ± 0.0158	0.5990 ± 0.0101	0.6512 ± 0.0109	0.7182 ± 0.0070
SC-VAR	0.2353 ± 0.0109	0.5703 ± 0.0077	0.8588 ± 0.0056	0.8928 ± 0.0035	0.7244 ± 0.0004	0.9276 ± 0.0088
SCRNN	0.6467 ± 0.0035	0.6617 ± 0.0047	0.6577 ± 0.0074	0.7245 ± 0.0249	0.7266 ± 0.0308	0.7644 ± 0.0297
Gain vs. best baseline	+0.5%	-28.4%	+33.9%	+8.6%	+2.9%	+9.3%

- **Take-home:** Water has no triangles, but STORM still improves medium and long horizons using node-edge topology.

Role of Topology

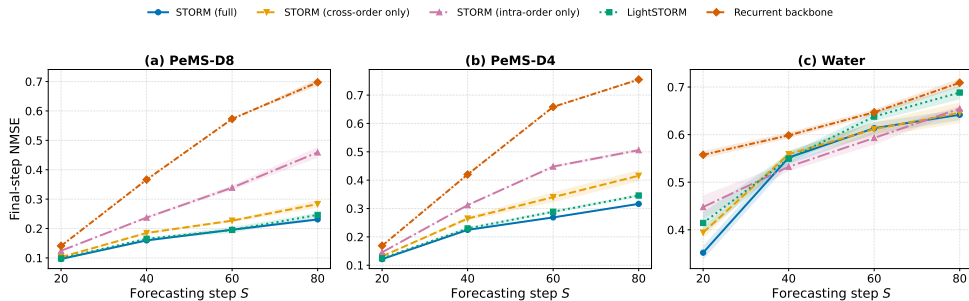


Figure: Topology ablation. Final-step NMSE at forecasting steps $S \in \{20, 40, 60, 80\}$. Lower is better. Full STORM combines intra-order Hodge filtering with cross-order boundary coupling.

- **Take-home:** topology-aware variants separate more clearly from the topology-free recurrent backbone at longer horizons.

Injection Modes: Results

Table: Injection-mode ablation on PeMS-D8. Lower is better. Horizontal separators distinguish input, hidden, and both injection modes.

Model	Mode	H	$S=10$	$S=40$	$S=80$
STORM	input	16	0.0616 ± 0.0006	0.2956 ± 0.0047	0.5655 ± 0.0088
	input	32	0.0587 ± 0.0007	0.2233 ± 0.0083	0.3913 ± 0.0053
	hidden	16	0.0578 ± 0.0014	0.1908 ± 0.0018	0.3256 ± 0.0200
	hidden	32	0.0560 ± 0.0004	0.1592 ± 0.0093	0.2370 ± 0.0125
	both	16	0.0573 ± 0.0001	0.1880 ± 0.0110	0.3376 ± 0.0098
	both	32	0.0541 ± 0.0007	0.1596 ± 0.0041	0.2310 ± 0.0059
LightSTORM	input	16	0.0636 ± 0.0004	0.3570 ± 0.0057	0.6919 ± 0.0100
	input	32	0.0626 ± 0.0008	0.3105 ± 0.0066	0.5867 ± 0.0046
	hidden	16	0.0588 ± 0.0001	0.2123 ± 0.0043	0.4065 ± 0.0064
	hidden	32	0.0552 ± 0.0008	0.1624 ± 0.0055	0.2511 ± 0.0177
	both	16	0.0600 ± 0.0006	0.2155 ± 0.0106	0.3736 ± 0.0104
	both	32	0.0558 ± 0.0006	0.1652 ± 0.0037	0.2462 ± 0.0067

Per-Order Results: PeMS-D4

Table: Per-order final-step NMSE at $S = 40$ and $S = 80$. Lower is better.

Model	$S=40$			$S=80$		
	Nodes	Edges	Triangles	Nodes	Edges	Triangles
STORM (both)	0.2318	0.2200	0.2030	0.3278	0.3120	<u>0.2433</u>
STORM (hidden)	<u>0.2324</u>	<u>0.2228</u>	0.1930	<u>0.3380</u>	<u>0.3264</u>	0.2535
LightSTORM (both)	0.2360	0.2265	<u>0.1990</u>	0.3560	0.3453	0.2377
LightSTORM (hidden)	0.2500	0.2432	0.2075	0.3769	0.3730	0.2803
iTransformer	0.2982	0.2878	0.3046	0.5307	0.5273	0.4904
xLSTM	0.4230	0.4105	0.4296	0.7504	0.7621	0.7615
GCN-xLSTM	0.3347	0.3840	0.4482	0.5863	0.6902	0.7532
SC-VAR	0.5827	0.6183	0.6822	0.9027	0.9619	0.9383

- **Take-home:** improvements are visible not only on nodes, but also on edge and triangle forecasts.

Per-Order Results: PeMS-D8

Table: Per-order final-step NMSE at $S = 40$ and $S = 80$. Lower is better.

Model	$S=40$			$S=80$		
	Nodes	Edges	Triangles	Nodes	Edges	Triangles
STORM (both)	<u>0.1600</u>	<u>0.1622</u>	0.1496	0.2272	0.2379	0.2153
STORM (hidden)	0.1598	0.1612	<u>0.1510</u>	<u>0.2316</u>	<u>0.2439</u>	<u>0.2244</u>
LightSTORM (both)	0.1630	0.1686	0.1584	0.2352	0.2569	<u>0.2244</u>
LightSTORM (hidden)	0.1609	0.1659	0.1532	0.2438	0.2627	0.2265
iTransformer	0.2440	0.2460	0.2443	0.4488	0.4742	0.4718
xLSTM	0.3466	0.3646	0.3681	0.6621	0.7303	0.7374
GCN-xLSTM	0.2371	0.3104	0.3391	0.4453	0.6063	0.6627
SC-VAR	0.5220	0.5840	0.6080	0.7946	0.9392	0.9483

- **Take-home:** at $S = 80$, STORM gives the strongest per-order results across nodes, edges, and triangles.

Further reading

- ▶ Topological signal processing and learning
 - ⇒ Barbarossa and Sardellitti, Topological Signal Processing over Simplicial Complexes, 2020
 - ⇒ Yang et al., Simplicial Convolutional Filters, 2022
 - ⇒ Yang et al., Simplicial Convolutional Neural Networks, 2022
 - ⇒ Krishnan et al., Simplicial Vector Autoregressive Models, 2024
 - ⇒ Mitchell et al., A Topological Deep Learning Framework for Neural Spike Decoding, 2024
 - ⇒ Jebali et al., STORM: Simplicial Topological Recurrent Models, under review
- ▶ Recurrent and spatio-temporal models
 - ⇒ Hochreiter and Schmidhuber, Long Short-Term Memory, 1997
 - ⇒ Beck et al., xLSTM: Extended Long Short-Term Memory, 2024
 - ⇒ Liu et al., iTransformer, 2023
- ▶ Theory
 - ⇒ Gama et al., Stability Properties of Graph Neural Networks, 2020
 - ⇒ Bartlett and Mendelson, Rademacher and Gaussian Complexities, 2002
 - ⇒ Mohri et al., Foundations of Machine Learning, 2018